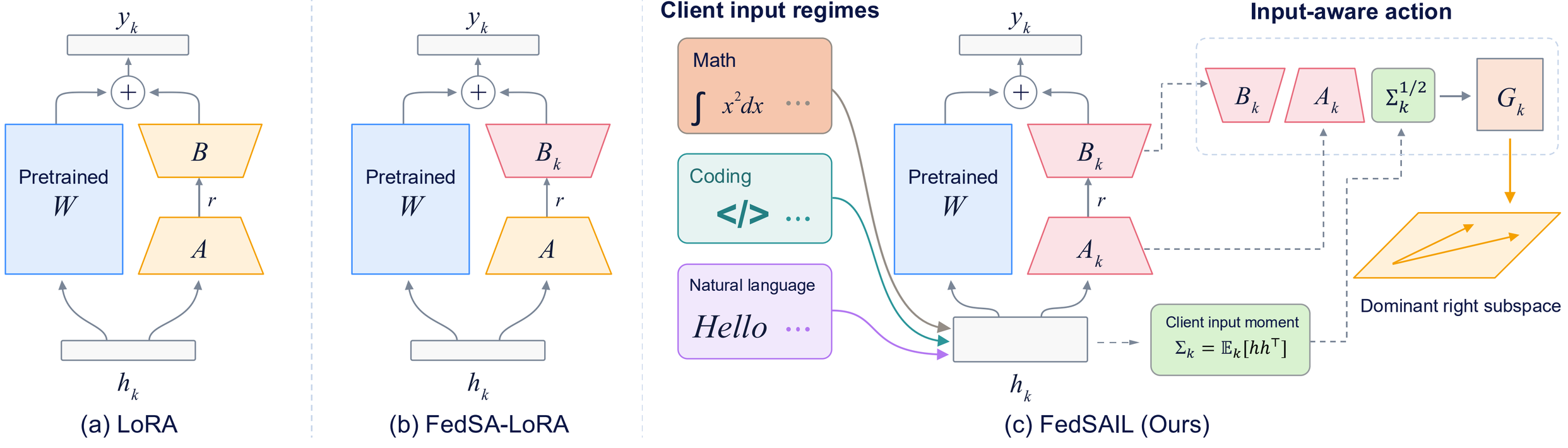

y_k
Pretrained
W
B
r
A
h_k
(a) LoRA
y_k
Pretrained
W
B_k
r
A
h_k
(b) FedSA-LoRA
Client input regimes
Math
$\int x^2 dx$ ...
Coding
</> ...
Natural language
Hello ...
y_k
Pretrained
W
B_k
r
A_k
h_k
Input-aware action
B_k
A_k
$\Sigma_k^{1/2}$
G_k
Dominant right subspace
Client input moment
$\Sigma_k = \mathbb{E}_k[hh^\top]$
(c) FedSAIL (Ours)
Frozen backbone
Shared object
Local object